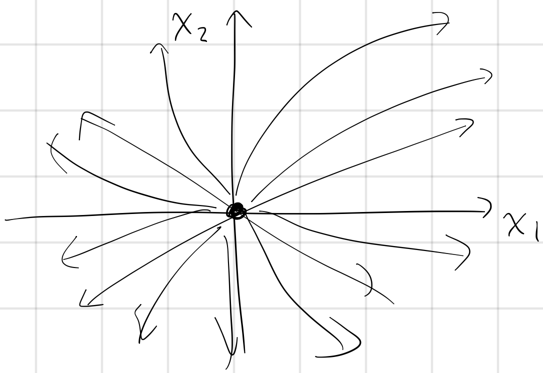
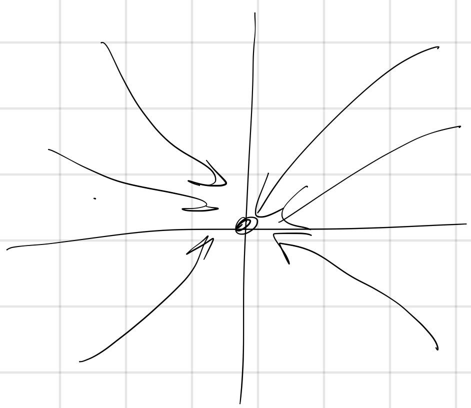
Stability

$$\frac{dx_1}{dt} = ax_1 + bx_2$$
$$\frac{dx_2}{dt} = cx_1 + dx_2$$

$$\Rightarrow \quad \frac{d\vec{x}}{dt} = A\vec{x} \qquad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$



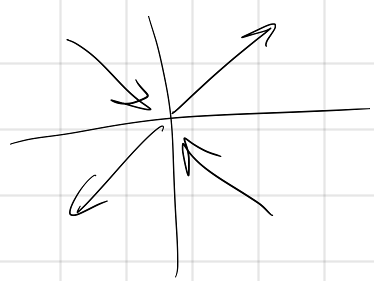
fixed point at $(0,0)$



stable - solutions go toward fixed point

unstable - not stable

(semi-stable/unstable)



The eigenvalues will be real

Determine stability and classify

29.) $A = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}$ triangular $\Rightarrow \lambda = 2, 3$

Stability: unstable

Classification: unstable node

32. $A = \begin{bmatrix} -5 & -2 \\ 6 & 3 \end{bmatrix}$ $\Delta = (-5)(3) - (-2)(6) = -15 + 12 = -3 < 0$

Stability: unstable

Classification: saddle point

$\det(A - \lambda I) = 0$

$\det\begin{pmatrix} -5-\lambda & -2 \\ 6 & 3-\lambda \end{pmatrix} = (-5-\lambda)(3-\lambda) - (-2)(6)$

$= -15 - 3\lambda + 5\lambda + \lambda^2 + 12$

$\lambda^2 + 2\lambda - 3$

$(\lambda - 1)(\lambda + 3) = 0$

$\lambda = 1, -3 \Rightarrow$ opposite signs

42. $A = \begin{bmatrix} 4 & -1 \\ 5 & -1 \end{bmatrix}$ $\Delta = 4(-1)-(5)(-1) = -4+5 = 1$

$\tau = 4 - 1 = 3$

Stability: unstable

Classification: unstable node

$\tau^2 - 4\Delta = (3)^2 - 4(1) = 9 - 4 = 5 > 0 \Rightarrow$ eigenvalues real

$\lambda^2 - \tau\lambda + \Delta = 0$

$\det\begin{pmatrix} 4-\lambda & -1 \\ 5 & -1-\lambda \end{pmatrix} = 0$

$\lambda^2 - 3\lambda + 1 = 0$

$(4-\lambda)(-1-\lambda) - (-1)(5) = 0$

$-4 + \lambda - 4\lambda + \lambda^2 + 5 = 0$

$\lambda^2 - 3\lambda + 1 = 0$

$\lambda = \dfrac{3 \pm \sqrt{5}}{2}$ $3 - \sqrt{5} > 0$

$3^2 > 5$

$9 > 5$

$\sqrt{9} > \sqrt{5}$

$3 > \sqrt{5}$

Assume imaginary

43. $A = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix}$ $\tau = 2 > 0$ $\Delta = 2(0) - (-1)(3) = 3$

Stability: unstable

Classification: Unstable spiral

$$\lambda = \frac{2 \pm \sqrt{2^2 - 4 \cdot 3}}{2}$$

$$= \frac{2 \pm \sqrt{4 - 12}}{2}$$

$$= \frac{2 \pm \sqrt{-8}}{2}$$

$$= \frac{2 \pm i\sqrt{8}}{2}$$

$$= \frac{2 \pm 2i\sqrt{2}}{2}$$

$$= \underbrace{1}_{>0} \pm i\sqrt{2}$$

46. $A = \begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix}$ $\tau = 2 > 0$ unstable spiral

56. $A = \begin{bmatrix} 2 & -3 \\ 3 & -2 \end{bmatrix}$ $\tau = 2 + (-2) = 0 \Rightarrow$ center

No assumptions about realness. Analyze stability and Classify

57. $A = \begin{bmatrix} -1 & -2 \\ 1 & 3 \end{bmatrix}$

$\Delta = (-1)(3) - (-2)(1)$
$= -3 + 2 = -1$

Stability: unstable

Classification: saddle node

59. $A = \begin{bmatrix} -1 & -1 \\ 5 & -3 \end{bmatrix}$

$\Delta = (-1)(-3) - (-1)(5)$
$= 3 + 5 = 8 > 0$

$\underbrace{\tau = -4}$, $\tau^2 - 4\Delta = (-4)^2 - 4 \cdot 8$
$= 16 - 32 = \underline{-16 < 0}$

$\boxed{\text{stable spiral}}$ imaginary

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

$$c_1 e^{\lambda_1 t}\vec{u} + c_2 e^{\lambda_2 t}\vec{v}$$